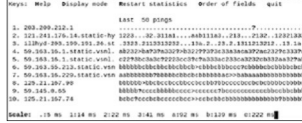

Figure 3: Display mode in mtr shows actual timing from the last 50 ping sequences

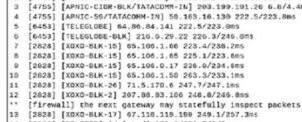

Figure 4: lft shows what's in between and who owns what

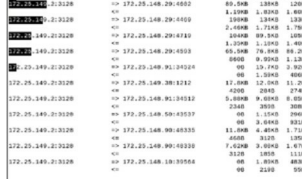

Figure 5: iftop -nNPB on a lightly-loaded system

is losing the most packets, as well as which ones are taking the most amount of time to reply. An even more useful display is obtained by hitting *j*, which shows you packet loss in absolute numbers instead of percentages. More importantly, it also shows you something called 'jitter', which means inconsistency in response times. You can also think of jitter as a measure of transient or occasional congestion in that link, causing only delays for now, though if the quality degrades further, there may be packet loss too. Seasoned travellers know that when too many flights show a 'delayed' status, sooner or later some will go from 'delayed' to 'cancelled'—this is pretty much the same thing.

The best feature of *mtr* can be seen by cycling the display mode (by pressing *d*). This is a very interesting display, showing the actual timing results from the last 50 ping sequences (or more, if your screen is wider). A "." means a reply was received, a ">" means it was received but took a long time, and a "?" means it has not been received yet. If you cycle the display mode again, the display changes to show 6 levels of granularity in the RTT, and a scale at the bottom to say what these levels mean. For example, in Figure 3, a "1" means a reply was received more than 5 but less than 14 ms later, and a ">" means a response packet was received more than 222 ms later.

This is a very cool display—and I can tell you from personal experience that it never fails to impress when you're trying to prove to someone that the problem is on *their* router! Most Windows-type admins are left speechless—although that's probably because they are trying to digest the fact that you don't need a bloated GUI framework to get useful work done!

Moving up a layer or two

However, it often happens that the system we are trying to trace does not accept ICMP (pings) or UDP (traceroute)—most security conscious admins disable everything that is not absolutely needed, and if it's a public Web server, it may only allow HTTP/HTTPS (ports 80/443). For times like this, you could just use the *traceroute*'s *-T* option, which uses TCP instead of UDP. It works pretty well, although this is not a continuously running program, so it tells you about connectivity and RTT for one round only.

However, we may want to find out who owns a particular network. When you need that, *lft* (Layer Four Trace) is pretty useful. Above and beyond what *traceroute* can do, *lft* can show if there are any firewalls in between, as well as what organisation owns those gateways or routers, as you can see in Figure 4.

Thinking local; *iftop* and *iptraf*

So let's say you've figured out who or what is slowing down your packets and (hopefully) got someone to fix it. Your traffic is moving pretty smoothly, and everyone is happy.

Actually, some people are too happy—they're hogging all the bandwidth! You need to find out who they are and have a quick word with them. The only question is: who is hitting the net so badly and what site are they hitting?

Even if you're not a Simon Travaglia, and you have only your own machine to worry about, perhaps you suddenly noticed a lot of activity on the network monitor (you do use one, right? I suggest *conky* for low-end machines and *gkrellm* for all others!) and you're wondering what program is doing it and why.

While *netstat* can certainly be used to give you